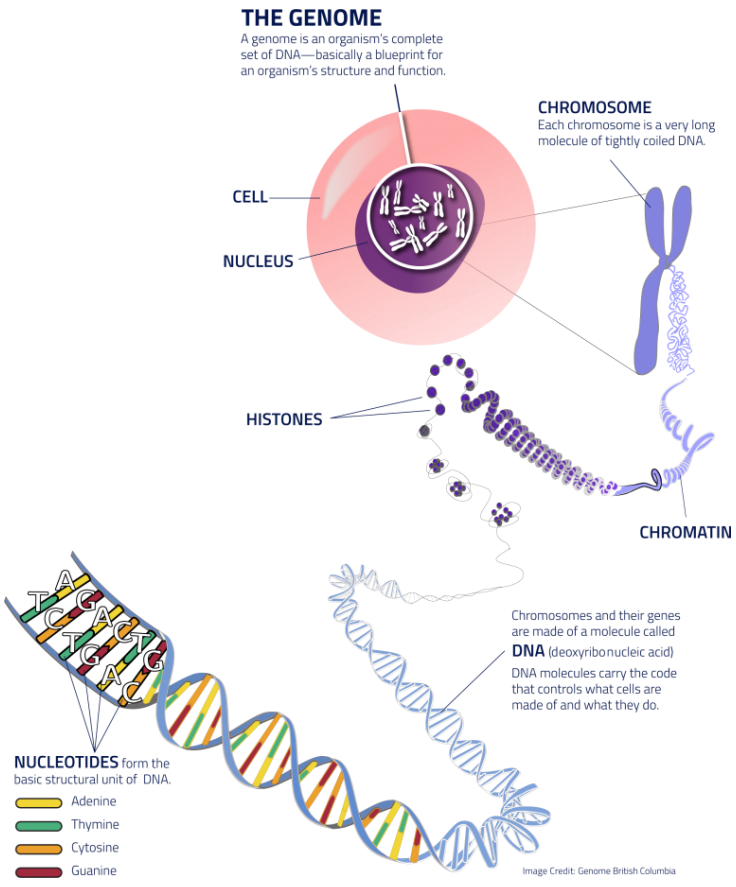
Computational Biology

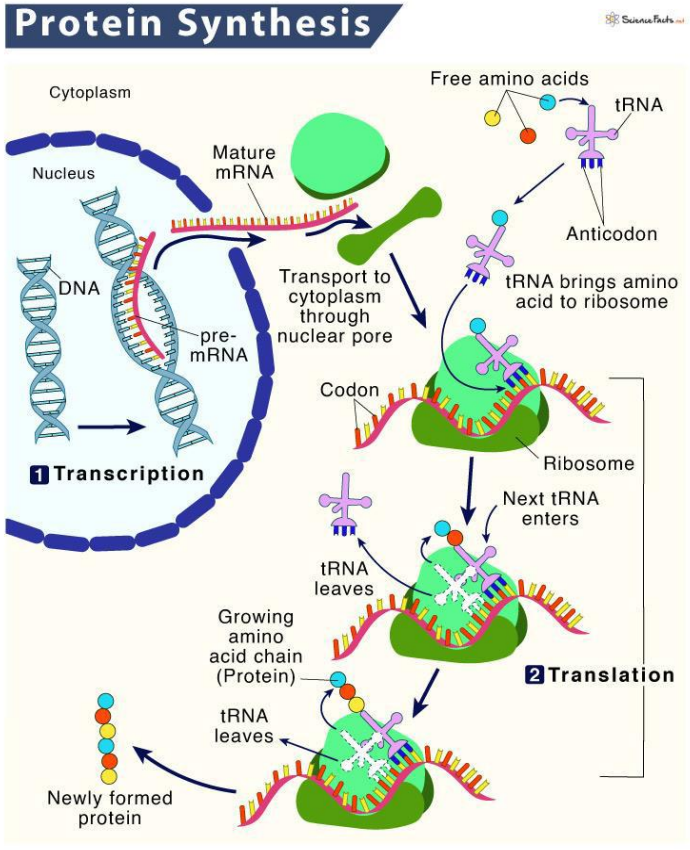
Session 6



Today's Agenda

- Single-Nucleotide Polymorphism
- Allele
- Genome-Wide Association Study





Review

- Humans have **46 chromosomes** (23 pairs) in most cells.
- The human genome contains about **3.2 billion DNA base pairs**.
- Humans have roughly **20,000–21,000 genes**, making up only **~1–2%** of all DNA.
- Almost every cell in the body contains the **same DNA**, but different genes are turned on or off.
- **Humans and chimpanzees** sharing approximately **98.6% to 99%** of their DNA sequences in direct base-pair comparisons.
- **Any two humans** are approximately **99.9%** genetically identical, meaning only 0.1% of DNA differs between any two individuals.

How to Read This Graphic

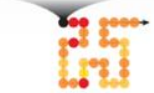
Each dot represents a sequence of about 500,000 pairs of chemical bases—the A, T, C and G of our genetic code—in the protein-coding portion of the human genome in the order that they appear on our chromosomes.

The color of the dot indicates how well the human sequence matches up with the corresponding sequence in the comparison species, with red signifying a greater difference between the two.

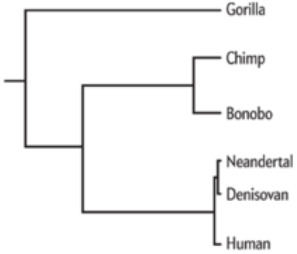
Fraction of different or unaligned bases (%)

0.2 0.5 1.5 3.0 6.0

ATGCCCGTTCTGAA ...



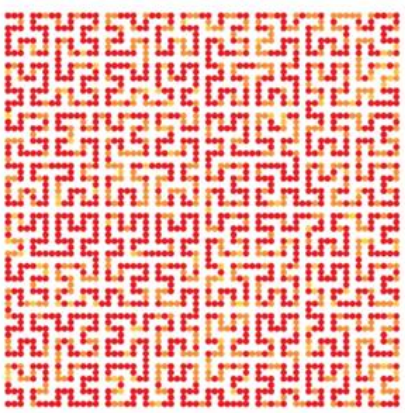
Family Tree



Graphic by Martin Krzywinski, Illustrations by Portia Sloan Rollings (for SCIENTIFIC AMERICAN*)*

SOURCES: UNIVERSITY OF CALIFORNIA, SANTA CRUZ, GENOME BROWSER (*chimp, gorilla and bonobo data*); MAX PLANCK SOCIETY (*Denisovan data*); "INITIAL SEQUENCE OF THE CHIMPANZEE GENOME AND COMPARISON WITH THE HUMAN GENOME," BY CHIMPANZEE SEQUENCING AND ANALYSIS CONSORTIUM, IN *NATURE*, VOL. 437; SEPTEMBER 1, 2005; "INSIGHTS INTO HOMINID EVOLUTION FROM THE GORILLA GENOME SEQUENCE," BY AYLWYN SCALLY ET AL., IN *NATURE*, VOL. 483; MARCH 8, 2012; "THE BONOBO GENOME COMPARED WITH THE CHIMPANZEE AND HUMAN GENOMES," BY KAY PRÜFER ET AL., IN *NATURE*, VOL. 486; JUNE 28, 2012; "A HIGH-COVERAGE GENOME SEQUENCE FROM AN ARCHAIC DENISOVAN INDIVIDUAL," BY MATTHIAS MEYER ET AL., IN *SCIENCE*, VOL. 338; OCTOBER 12, 2012

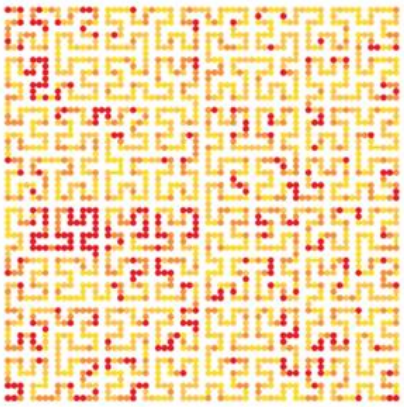
https://www.scientificamerican.com/article/tiny-genetic-differences-between-humans-and-other-primates-pervade-the-genome/





Gorilla

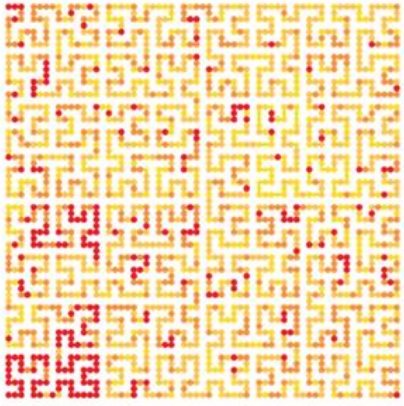
On the whole, our coding genome differs more from the gorilla's than from the chimp's or the bonobo's, reflecting the fact that we have been evolving along separate trajectories for a longer period. But about 15 percent of the human genome looks more like the gorilla's than the chimp's or the bonobo's.





Bonobo

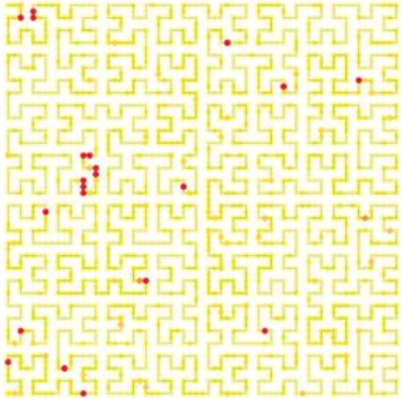
The genome of the bonobo—which has a social structure centered on females—shows it to be just as closely related to us as chimps are, although we differ from the two species in distinctive ways. These findings may force scientists to reconsider how our long-ago forerunners lived.





Chimp

Researchers have traditionally considered the chimpanzee, which lives in patriarchal societies, to be our closest living relative and thus the best model for reconstructing the lives of ancient human ancestors. The recent genome-sequencing work calls that view into question, however.

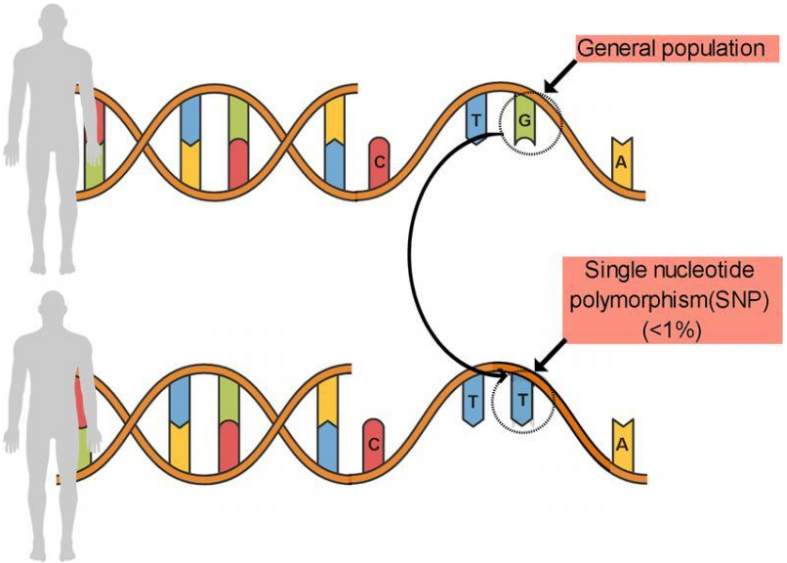




Denisovan

The Denisovans—a group of archaic humans closely related to the Neandertals—show far fewer sequence differences from us than any of the African apes do, having shared a common ancestor with *H. sapiens* in the much more recent past, around 400,000 years ago.

Single-Nucleotide Polymorphism



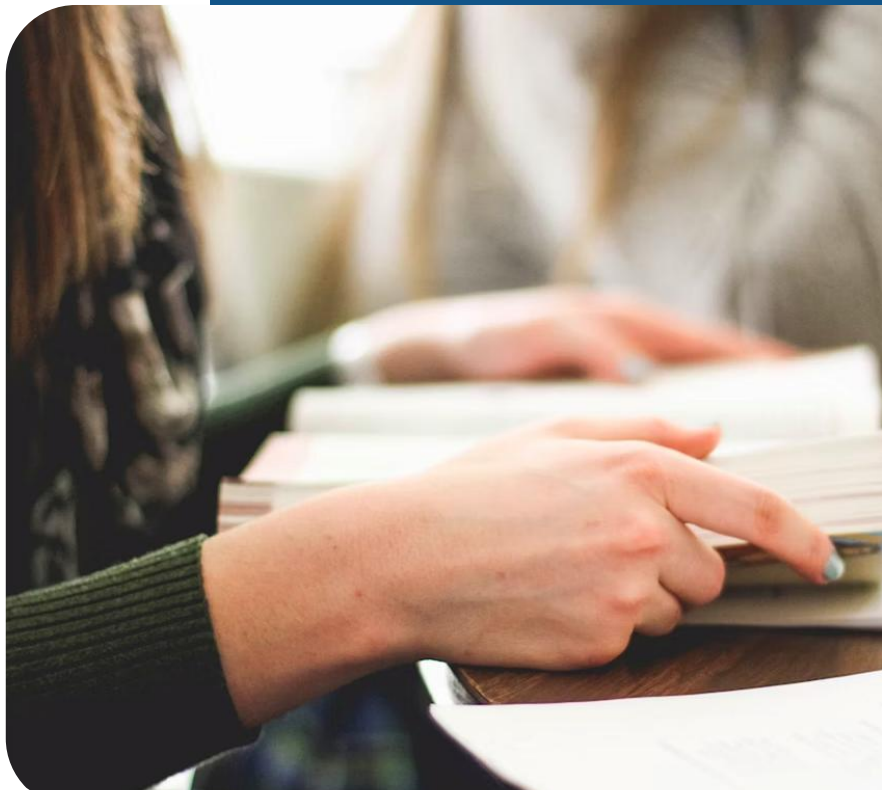
In genetics and bioinformatics, a single-nucleotide polymorphism, SNP(s), is a germline substitution of a single nucleotide at a specific position in the genome. Although certain definitions require the substitution to be present in a sufficiently large fraction of the population (e.g. 1% or more).

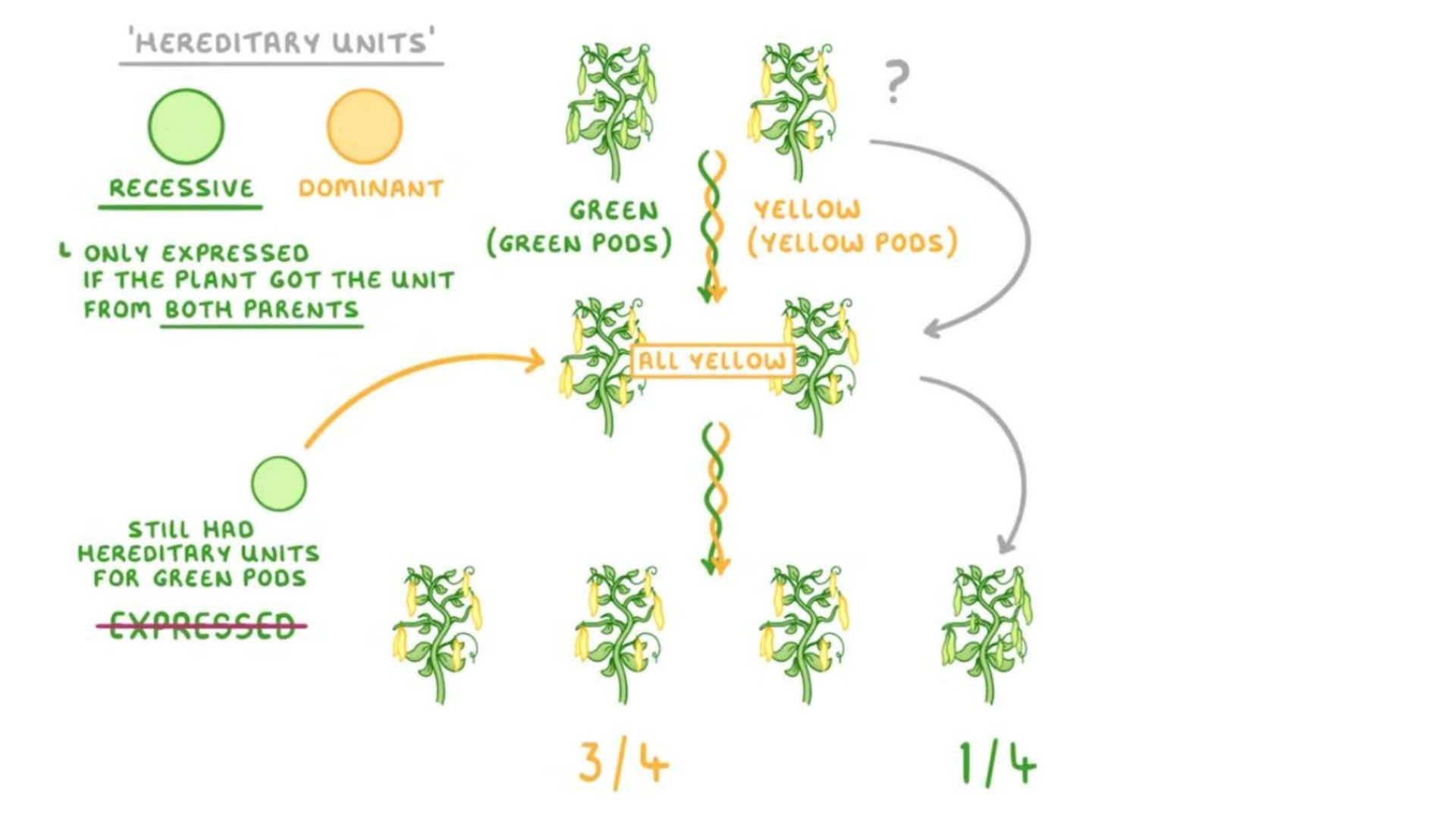
For example, a G nucleotide present at a specific location in a reference genome may be replaced by an A in a minority of individuals. The two possible nucleotide variations of this SNP – G or A – are called **alleles**.

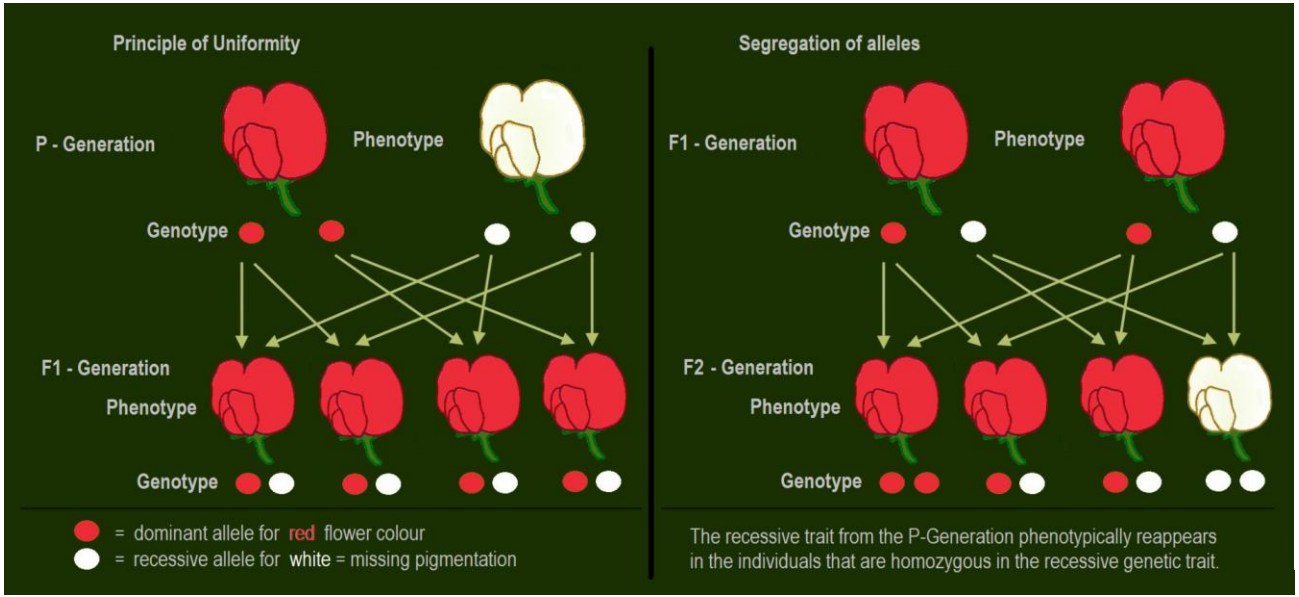


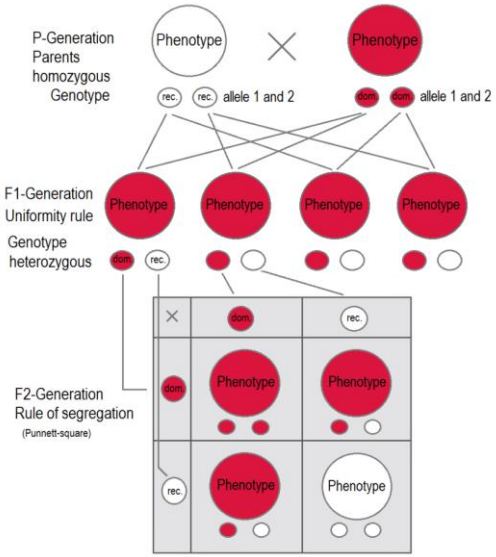
Gregor Mendel

Mendelian inheritance

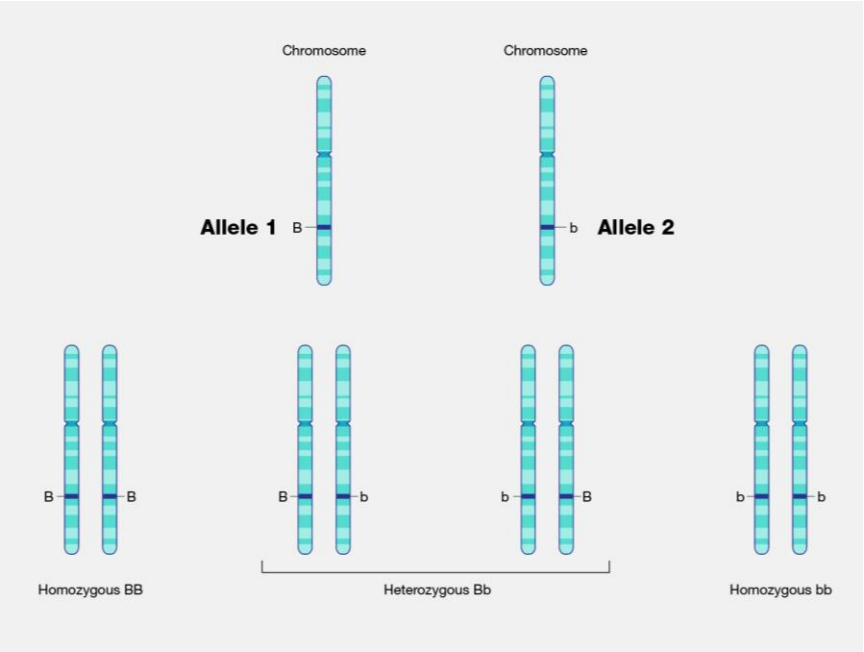



'HEREDITARY UNITS'
RECESSIVE
DOMINANT
ONLY EXPRESSED IF THE PLANT GOT THE UNIT FROM BOTH PARENTS
GREEN (GREEN PODS)
YELLOW (YELLOW PODS)
?
ALL YELLOW
STILL HAD HEREDITARY UNITS FOR GREEN PODS
EXPRESSED
3/4
1/4


Principle of Uniformity
P - Generation
Phenotype
Genotype
F1 - Generation
Phenotype
Genotype
= dominant allele for red flower colour
= recessive allele for white = missing pigmentation
Segregation of alleles
F1 - Generation
Phenotype
Genotype
F2 - Generation
Phenotype
Genotype
The recessive trait from the P-Generation phenotypically reappears in the individuals that are homozygous in the recessive genetic trait.

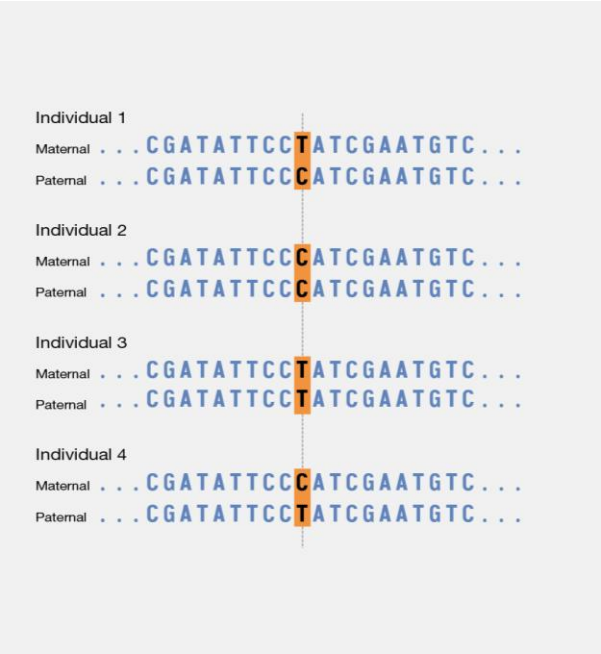

P-Generation
Parents
homozygous
Genotype
Phenotype
Phenotype
rec.
rec.
allele 1 and 2
dom
dom
allele 1 and 2
F1-Generation
Uniformity rule
Genotype
heterozygous
Phenotype
Phenotype
Phenotype
Phenotype
dom
rec.
F2-Generation
Rule of segregation
(Punnett-square)
×
dom
rec.
dom
Phenotype
Phenotype
rec.
Phenotype
Phenotype

Allele



An allele is one of two or more versions of DNA sequence (a single base or a segment of bases) at a given genomic location. An individual inherits two alleles, one from each parent, for any given genomic location where such variation exists. If the two alleles are the same, the individual is homozygous for that allele. If the alleles are different, the individual is heterozygous.

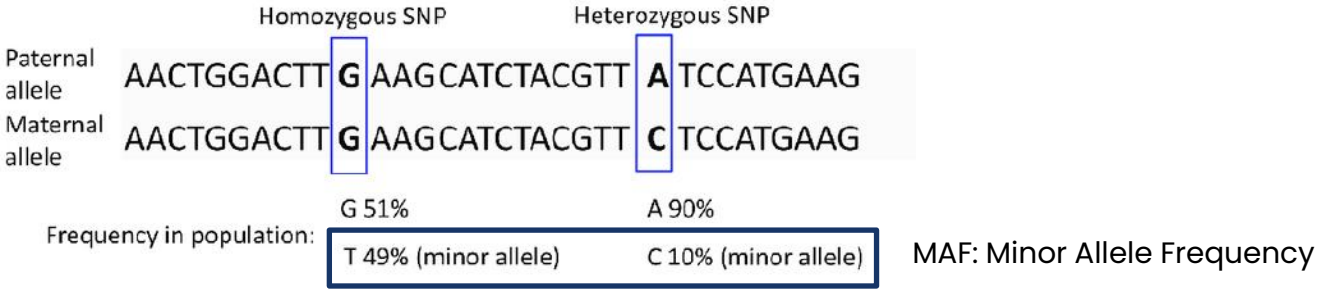
Single-Nucleotide Polymorphism



A single nucleotide polymorphism (abbreviated SNP, pronounced snip) is a genomic variant at a single base position in the DNA. Scientists study if and how SNPs in a genome influence health, disease, drug response and other traits.

Example SNP Catalog:
https://www.ebi.ac.uk/gwas/variants/rs7903146

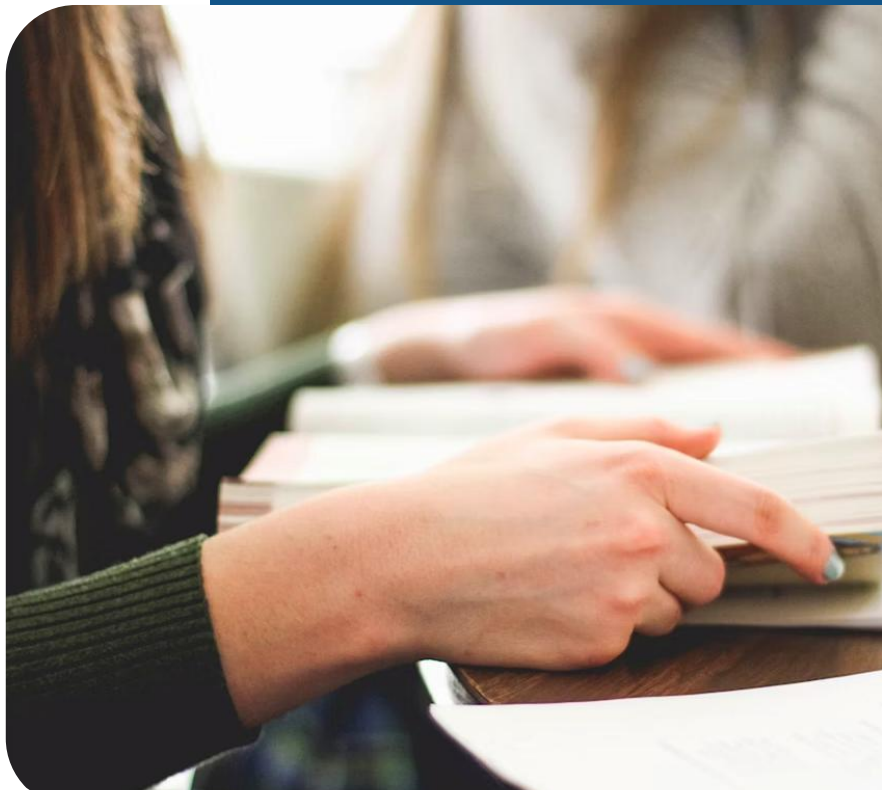
Single-Nucleotide Polymorphism



MAF: Minor Allele Frequency

- Values range from 0 – 2. But genotype imputation may result in decimal values.
- Homozygous major SNPs are marked as 0, homozygous minor as 2, and heterozygous as 1.

Genome-Wide Association Study

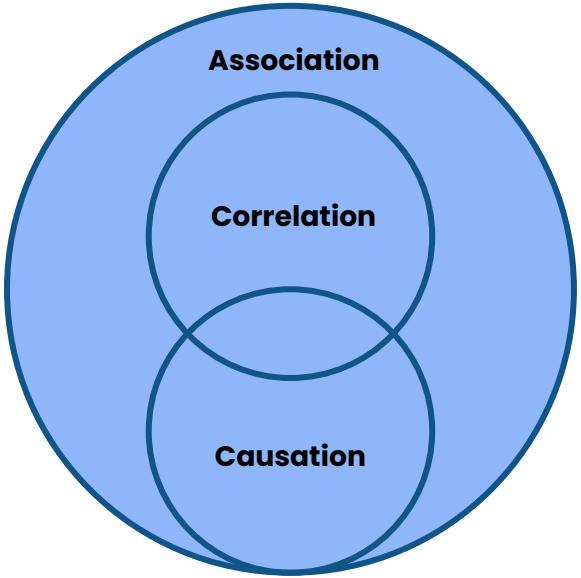


Association-Correlation-Causation

Correlation is a statistical measure that expresses the extent to which two variables are *__linearly__ related (meaning they change together at a constant rate).

Correlation **does not** mean causation!
Causation is the relationship where one variable **affects** another

Association is any relationship between 2 variables, whether it's linear or non-linear



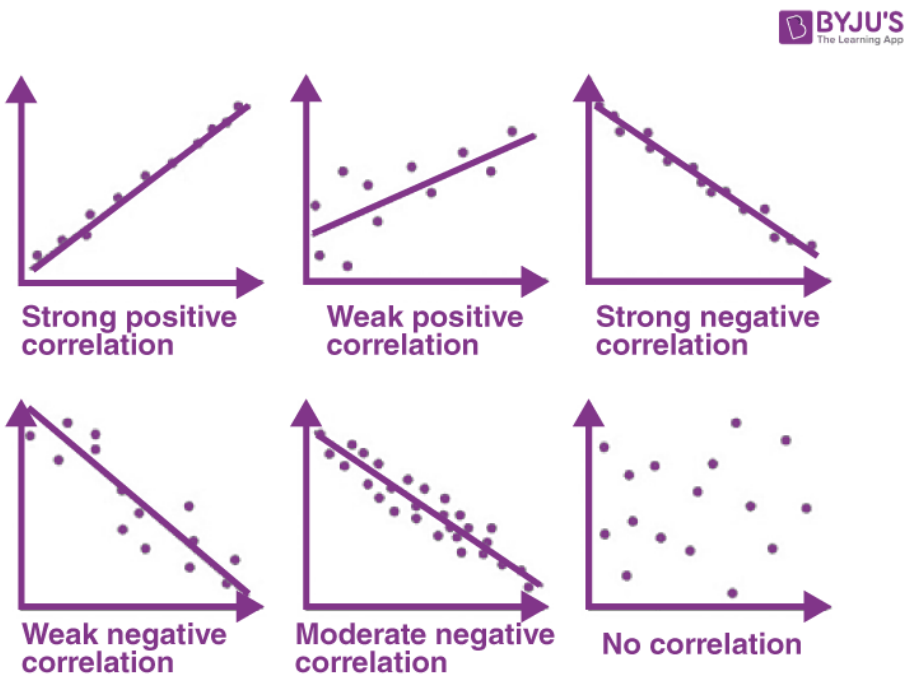
*With some exception

CORRELATION

Correlation is a statistical measure that expresses the extent to which two variables are linearly related (meaning they change together at a constant rate).

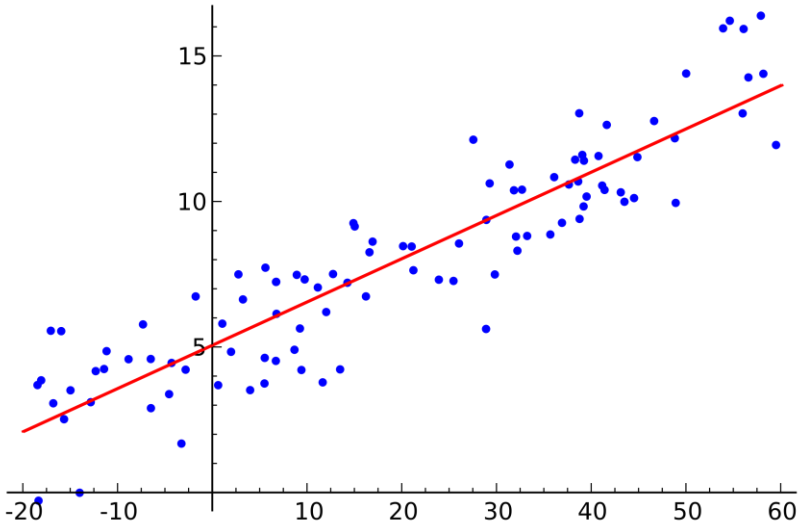
Correlation coefficient:

$$-1 \leq r \leq 1$$



LINEAR REGRESSION

Linear regression is a linear approach for **modelling** the **relationship between** a scalar response and one or more explanatory variables (also known as **dependent and independent variables**).
Linear regression:
$Y = b_0 + b_1 X$



Association Study

Is ‘Weekly study duration’ associated with ‘GPA’? ➔ Correlation
Is ‘IQ’ associated with ‘GPA’? ➔ Correlation
Is ‘Gender’ associated with ‘GPA’? ➔ Two sample testing

How ‘Weekly study duration’ affects ‘GPA’? ➔ Multinomial Regression
How ‘IQ’ affects ‘GPA’?
How ‘Gender’ affects ‘GPA’?

$$Y = b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3$$

Association Study

$Y = b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3$
Y = 1.2 + 0.2 Study hour + 0.007 IQ + 0.0003 Gender (F=1, M=0) (this is arbitrary number)

Female with IQ of 110, study 10 hours a week:
$Y = 1.2 + 0.2 \times 10 + 0.007 \times 110 + 0.0003 \times 1 = 1.2 + 2 + 0.77 + 0.0003 = 3.9703$

Male with IQ of 150, study 1 hours a week:
$Y = 1.2 + 0.2 \times 1 + 0.007 \times 150 + 0.0003 \times 0 = 1.2 + 0.2 + 1.05 + 0 = 2.45$

Significancy (p-value):

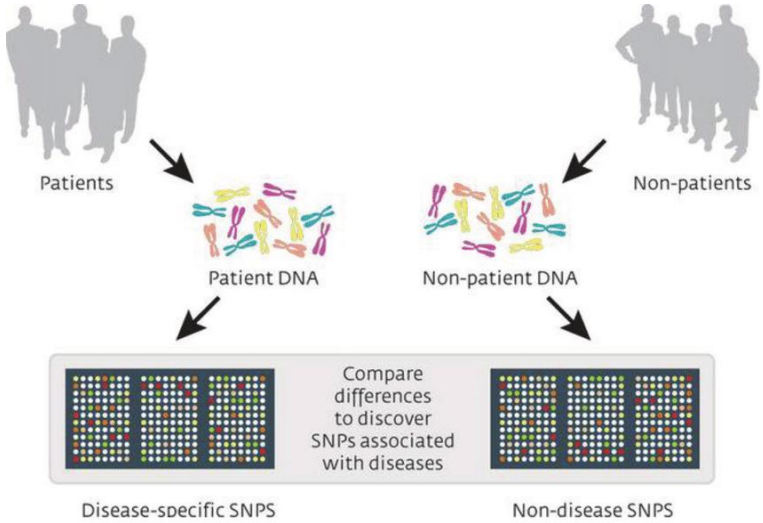
Study hour	: 0.00003 (significant)
IQ	: 0.008 (significant)
Gender	: 0.3 (not significant)

Try this on Genetic level?
$Y = b_0 + b_1 X_1 + b_2 X_2 + b_3 X_3 \ldots + b_{3.200.000.000} X_{3.200.000.000} + b_{3.200.000.001} X_{3.200.000.001}$
Y = ?

Genome Wide Association Study

Genome-wide association studies (GWAS) are **observational** studies that scan the entire genomes of large populations to identify genetic variants—typically Single Nucleotide Polymorphisms (SNPs)—**statistically associated with specific diseases, traits, or conditions**. By comparing DNA from affected individuals (cases) with healthy ones (controls), researchers can identify common genetic risk factors.

The purpose of GWAS is to understand the genetic basis of common, complex diseases (e.g., Type 2 diabetes, cancer, Alzheimer's) and traits to aid in prediction, prevention, and treatment.

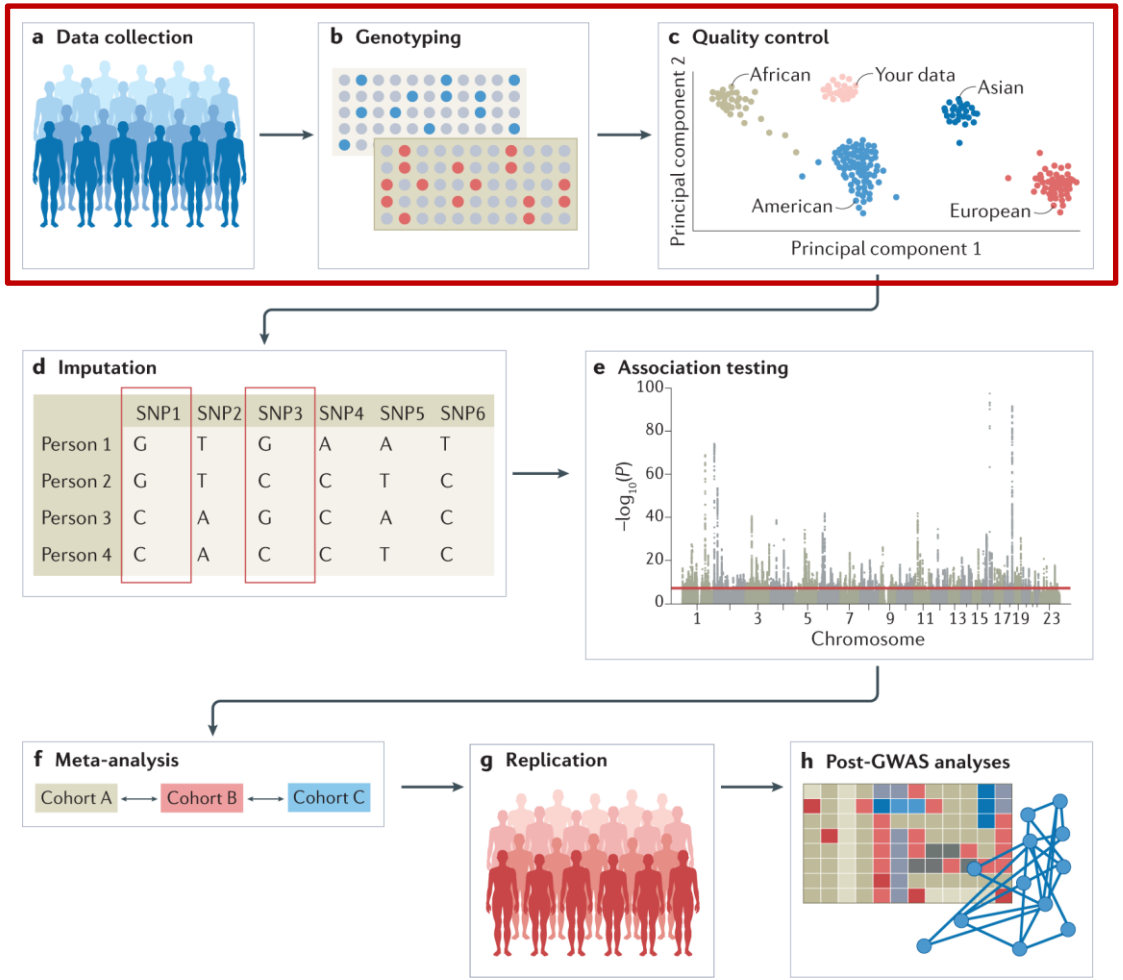


Method

a. Data collection
- Collection of genotype, clinical, & phenotype data.
- Sample recruitment should not be biased.

b. Genotyping
- Collection of genotype data.
- Uses microarray, or NGS methods (WGS and WES).

c. Quality control
- Both on wet and dry labs.
- Include genotype calling, missing SNP checking, MAF calibration, heterozygosity rate evaluation, etc.
- Dimensionality reduction can be utilized here.



https://doi.org/10.1038/s43586-021-00056-9

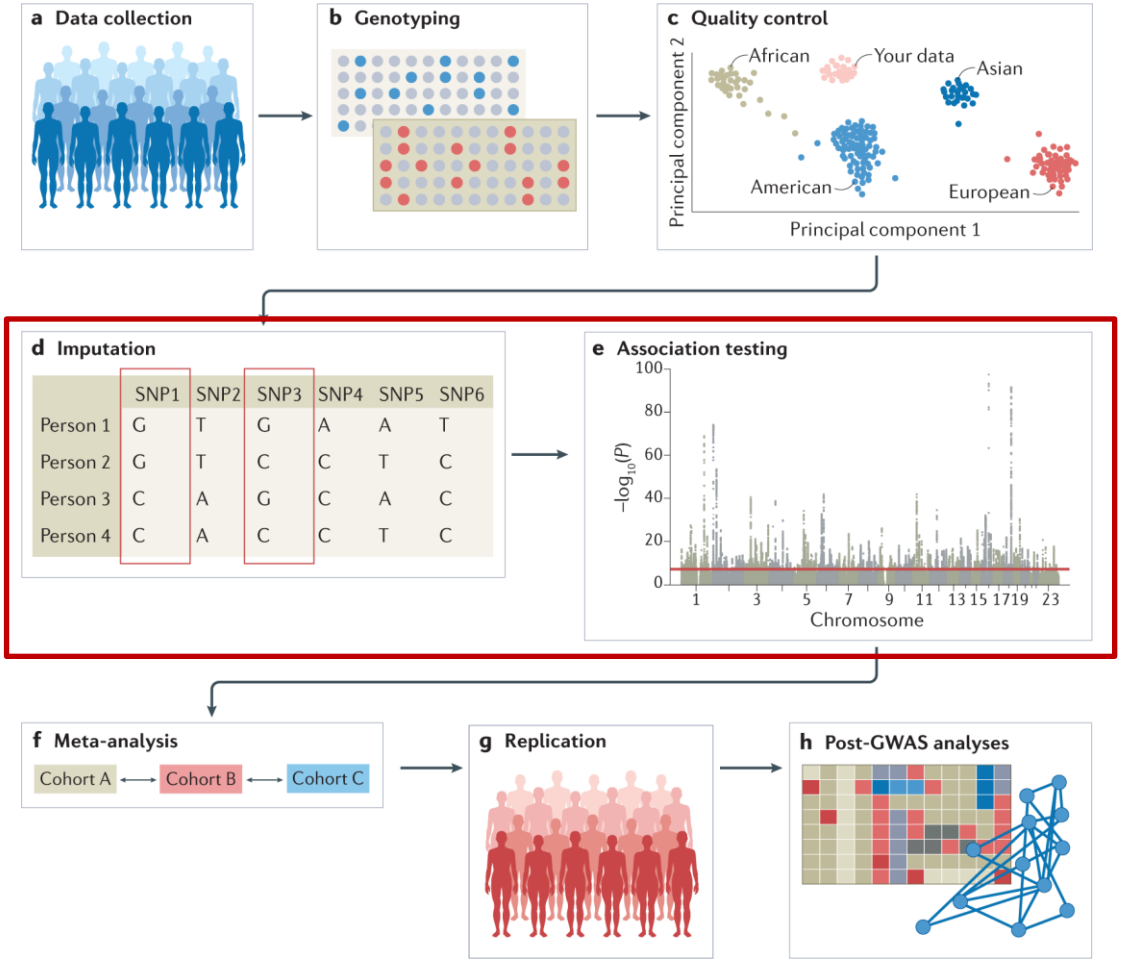
Method

d. Imputation

- Filling of missing SNP values.
- Can be performed on sequence-level or genotype-level.
- On sequence-level, imputation can be based on the reference population genome.

e. Association testing

- Run statistical or regression tests for each variant.
- Summary statistics are analyzed.
- Confounders and covariates should also be accounted for



https://doi.org/10.1038/s43586-021-00056-9

Method

f. Meta-analysis

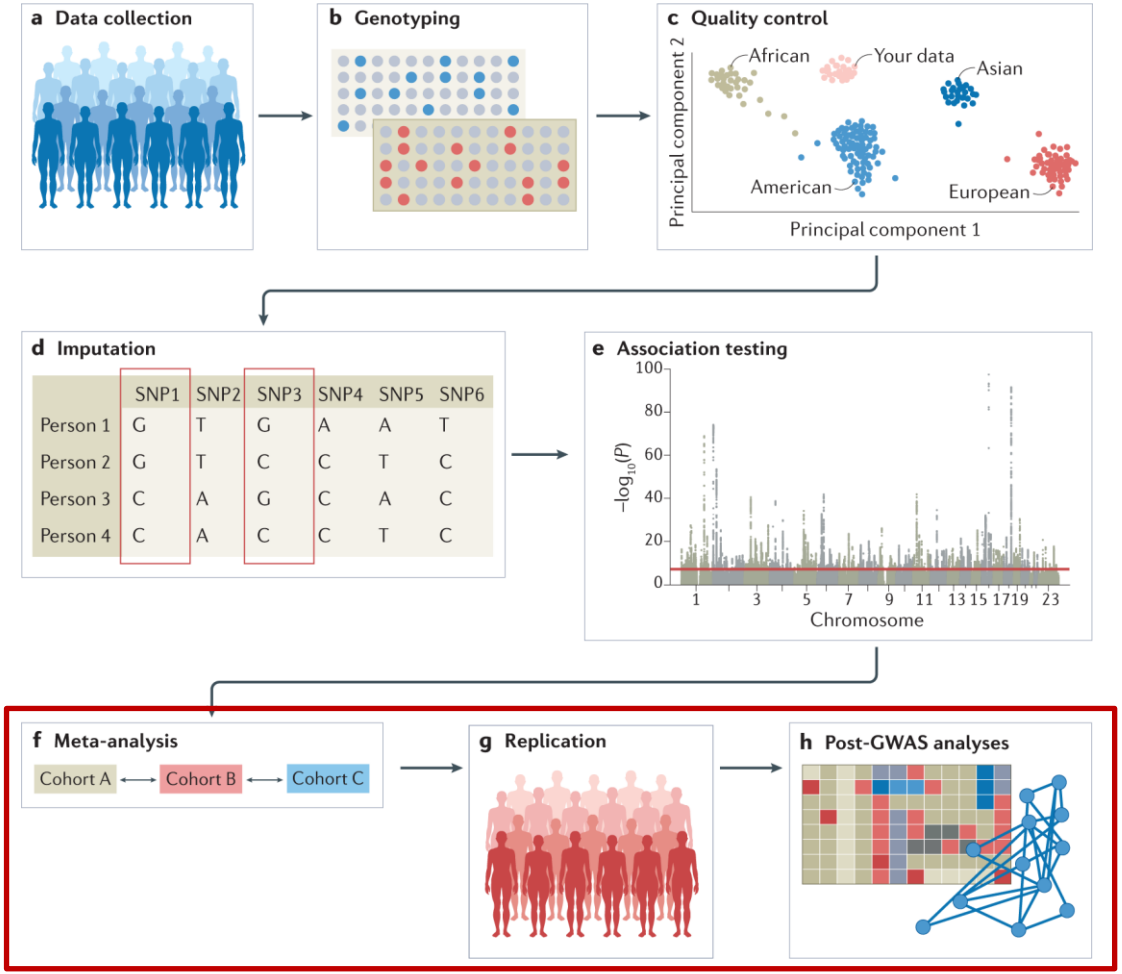
- Combine results from smaller cohorts using standardized pipelines.

g. Replication

- Replicate the experiment on cohorts.

h. Post-GWAS analyses

- Further analysis from discovered associations using external resources.
- Examples: SNP-to-gene mapping, in-silico fine-mapping, gene-environment interaction analysis, polygenic risk predictions, etc.

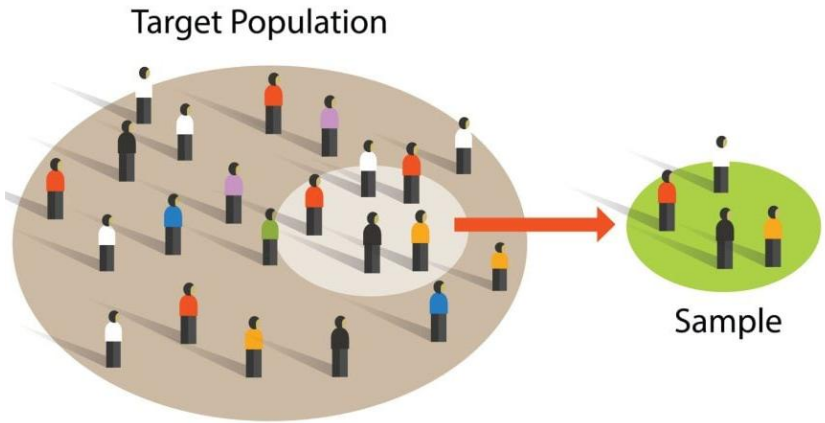


https://doi.org/10.1038/s43586-021-00056-9

Data Collection

Case-control Proportion:
Number of case and control samples should be equal or close to equal to prevent errors caused by data imbalance.

Confounders and covariates:
Confounders, Variables that are usually associated with genotype and phenotype but are usually not regarded as a cause to the phenotype.
Ex: age, sex, ethnicity, life style.
Covariates, confounders included in the GWAS regression model to account for confounding factors.
Ex: age, sex, smoking status.



Quality Control

Missingness Rate:
- SNPs with too many missing values should be removed.
- Common threshold is 5% missing rate.
- SNPs with different missingness rate between case and control samples should also be removed as it may cause errors.

Minor Allele Frequency (MAF):
- If the MAF is too small, the SNP may be random occurrences, not valid genetic variants.
- Common threshold is 1%.

Hardy-Weinberg Equilibrium
- Describes the expected distribution of alleles.
- Lack of heterozygous alleles may result in such cases.

p = frequency of allele A
q = frequency of allele a

$$p + q = 1$$

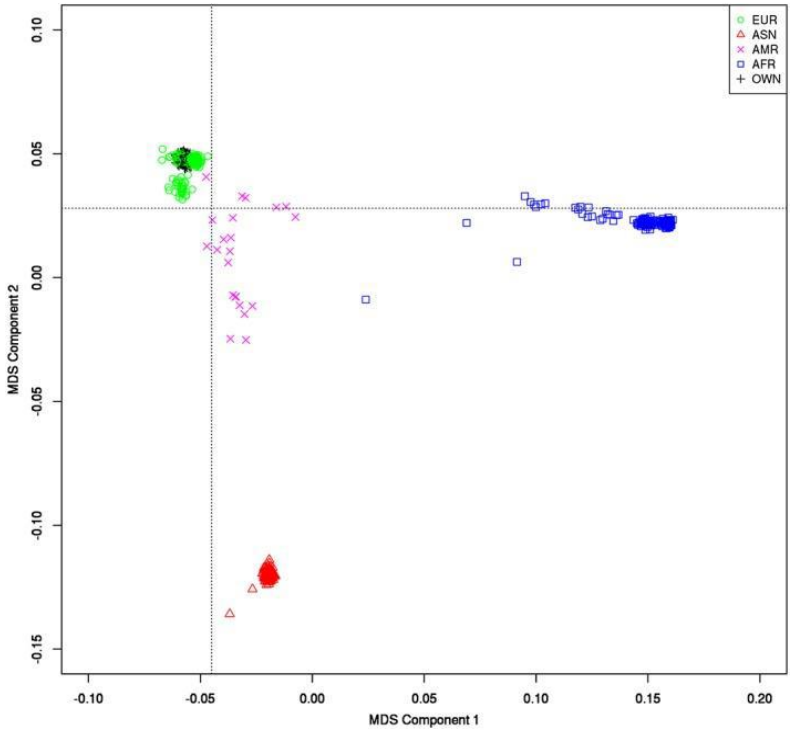
$$(p + q)^2 = 1$$
$$p^2 + 2pq + q^2 = 1$$

p^2 = frequency of genotype AA
$2pq$ = frequency of genotype Aa
q^2 = frequency of genotype aa

Quality Control

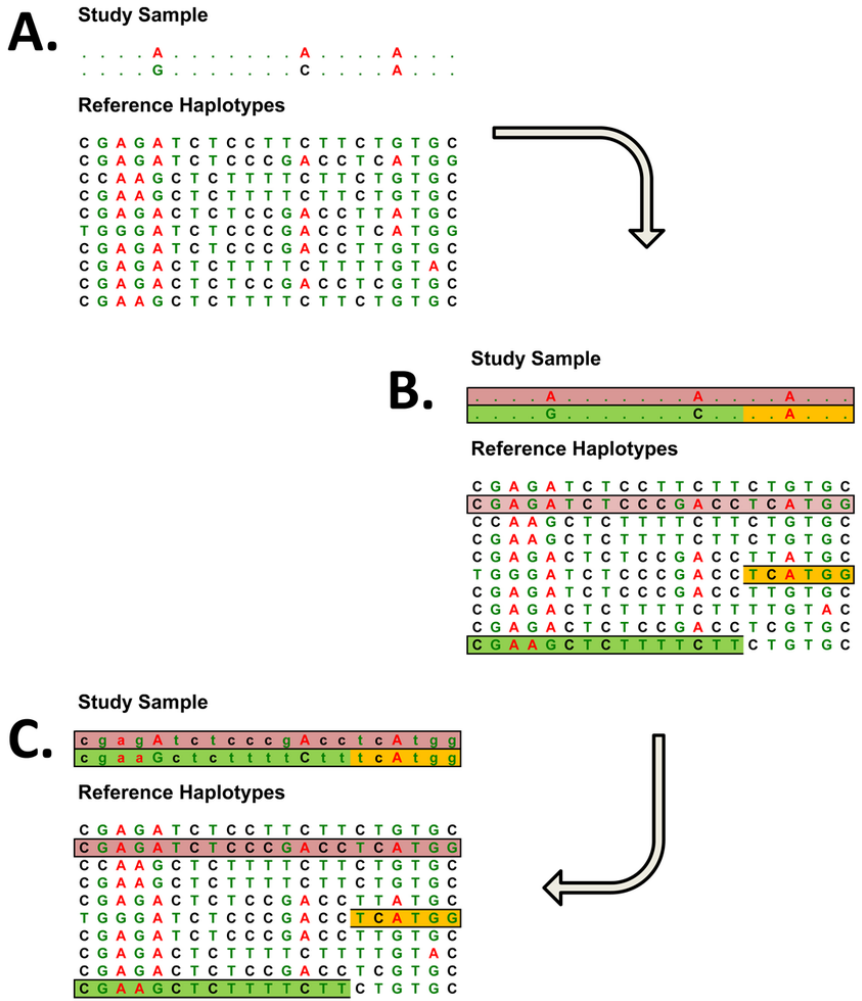
Population Stratification:

- Differences on allele frequencies may occur between subpopulations.
- Genetic variants can be correlated by coincidence to population structures.
- Commonly used as covariates.
- Methods to handle such cases:
  - Splitting the analysis by subpopulations.
  - PCA
- Multidimensional scaling



Imputation

- Infer missing genotype values using statistical estimations. This estimation is known as haplotype estimation.
- Estimation methods vary from multinomial model, Hidden Markov Model (HMM), PHASE, etc.
- Tools: MaCH Minimac, IMPUTE2, Beagle, Michigan Imputation Server, etc.



Statistical Analysis

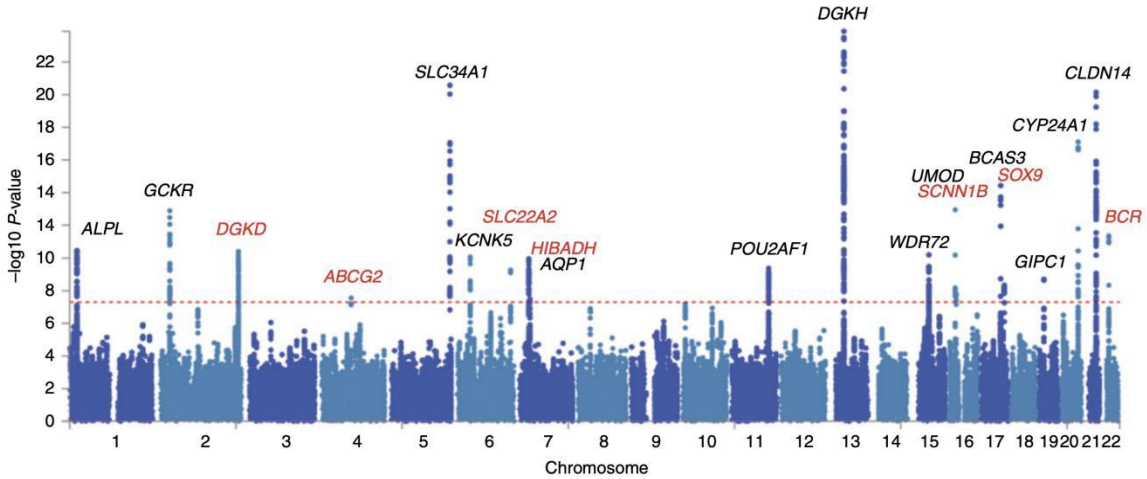
- Common formula: logistic regression, or Additive model for linear regression cases
- Other methods: chi-squared test
- Variables of interests: p-values, odd-ratios
- The p-value threshold in GWAS analysis is **5e-8**.

Iteration for all SNP

$$Y = b_0 + b_{SNP}\,SNP + b_1\,Cov1 + b_2\,Cov2 + \ldots + b_n\,Cov_n$$

$$Y = b_0 + b_{SNPA}\,SNP_A + b_1\,Cov1 + b_2\,Cov2 + \ldots + b_n\,Cov_n$$
$$Y = b_0 + b_{SNPB}\,SNP_B + b_1\,Cov1 + b_2\,Cov2 + \ldots + b_n\,Cov_n$$
…
…
$$Y = b_0 + b_{SNP3.2B}\,SNP_{3.2B} + b_1\,Cov1 + b_2\,Cov2 + \ldots + b_n\,Cov_n$$

Interpretation



- Commonly used plot to visualize SNP associations with the phenotype.
- SNP ID are placed on the x-axis while –log(p-value) is the y-axis
- SNPs with the strongest association will have visibly distinct negative log value.
- This example is taken from a GWA study investigating kidney stone disease, so the peaks indicate genetic variants that are found more often in individuals with kidney stones.

THANK YOU!